

Gabriella Shepherd

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EDUCATION

Bachelor of Science, Mechanical Engineering, Dec 2028

The University of Texas at Austin | GPA: 3.51/4.00

Certificate in Core Texts and Ideas; Jefferson Scholars Program

Relevant Coursework: Dynamics, Mechanics of Solids, Engineering Computational Methods, Thermodynamics

ENGINEERING EXPERIENCE

Lead Drivetrain Engineer, Longhorn Racing (Formula SAE Electric), Sept 2025 – Present

- Leading drivetrain architecture, design and vehicle integration for the 2026-27 FSAE electric vehicle
- Developed four conceptual architectures in CAD to evaluate mass, CG, load paths, suspension/frame/battery integration, and serviceability; prioritized reducing complexity by deleting parts
- Selected architecture around team target of 56% rear biased longitudinal car CG, targeting reduced frame weight through a smaller rear box, deleted rear impact structure tubes, and minimized need for drivetrain reinforcement tubes
- Independently learned AC TIG welding; welded Motor Scatter Shield and Jacking Point; machined battery mounts

PROJECTS

Stub Shafts | SolidWorks, ANSYS Mechanical, Excel | Longhorn Racing | Oct 2025 – Dec 2025

- Designed custom Stub Shafts to withstand nearly **1000 Nm of torque**, achieving an **8% mass reduction**
- **Validated over 275 miles** of testing with no observed fatigue failures
- Optimized fatigue performance by analyzing critical load cases, Marin factors, and stress concentrations at the spline-shaft transition across a nearly 600,000 cycle design lifespan
- Developed Excel spreadsheet to calculate **Marin factors**, angular deflection, and **Hertzian contact stress**
- Modeled tripod contact stress and worst-case halfshaft angle in **ANSYS Mechanical** to identify stress concentrations
- Delivered **GD&T drawing** with attention to spline callout table, total runout over shaft and perpendicularity to ensure seamless interfacing with COTS differential geometry
- Researched material properties and selected optimal steel grade, heat treatment, and finish

Jacking Point | SolidWorks, ANSYS Mechanical | Longhorn Racing | Dec 2025 – Feb 2026

- Designed vehicle Jacking Point to seamlessly interface with existing drivetrain/dynamics assemblies
- Validated M4 bolts for **2.3 kN shear loads** via hand calculations
- Designed for manufacturing by optimizing geometry for 2D laser-cutting and **weldability**
- Built a **strategically meshed ANSYS simulation** that enabled quick iteration, **load-path driven pocketing**, and analysis of worst-case load conditions to prevent excessive stress and buckling
- Implemented parametric CAD updates across part and greater drivetrain assembly

ADDITIONAL EXPERIENCE

Founder and Business Owner, Byellafait, Apr 2020 – Jun 2023

- Amassed more than 590 sales and over 160 five-star reviews in 12 countries selling original artwork
- Self-taught Search Engine Optimization, marketing, photo/video production, manufacturing, Adobe Suite

SKILLS

Technical: SolidWorks (CAD), ANSYS Mechanical (FEA), GD&T/Engineering Drawings, DFM, Machining (Manual Mill/Basic Shop Tools), TIG Welding, Excel, Python, Java

Certifications: CPR, Basic Life Support (BLS), Emergency Oxygen, First Aid and Lifeguard